Tumor Segmentation Using Neural Networks

1st Tim Perr
Computer Science/Data Science
Michigan Technological University
Michigan, USA
tlperr@mtu.edu

2nd Sam Weiss
Data Science
Michigan Technological University
Michigan, USA
saweiss@mtu.edu

3rd Alex Gore
Computer Science
Michigan Technological University
Michigan, USA
algore@mtu.edu

4th Tony Garnett
Computer Science
Michigan Technological University
Michigan, USA
tmgarnet@mtu.edu

I. Introduction and Motivation

Healthcare is a daunting entity that everyone must think about at some point in their life. Whether that be for minor illnesses such as a light cough, or more major sicknesses such as cancer, everyone must consider it. Specifically, cancer has been an illness that has historically been an uphill battle to fight and find cures for. Having efficient and accurate modes to diagnose and treat such an ailment early is of upmost importance, and in some cases it can be a life or death situation.

In the worst case scenario, extreme cases such as brain cancer have low 5-year survival rates. For patients who have a glioblastoma multiforme tumor, survival rates are 14% for patients aged 20-44, 4% if the patient is 45-54, and 1% for patients 55-64 [1]. Improving these rates is a goal that many people are working toward. One of the ways we can do this with early detection, which can improve a patients overall outlook, and possibly make the 5-year survival rates higher by identifying and removing tumors before they spread to rapidly.

One such example of early detection that we can do is via machine learning methods. Some methods, such as using neural networks, can help doctors identify tumors possibly before they would be able to identify it themselves. Upon successful implementation of our methods, doctors could be able to use our models to aid in their diagnosing, and therefore save lives.

II. Literature Review

Tumor segmentation is a process that many have attempted with various strategies and methods, each of which have yielded differing results [2], [3]. These methods range from simple image processing algorithms to deep learning approaches, of which neural networks have been a proven concept to achieve significantly successful results while also being highly efficient [2], [3].

Gordillo et al. discussed different methods for the segmentation of brain tumors. They were able to conclude that it is important to address the segmentation towards a fully automated method. They also noted that it would be desirable to have unsupervised fully automatic segmentation method in order to avoid the use of patient-specific training. Additionally, the use of some pre- or post- processing methods was discussed as a way to provide more reasonable segmentation results [2].

A literature review by Wadhwa et al. of different methods of brain tumor segmentation from brain MRI images. They included the performance and quantitative analysis of different methods. Their review showed that ensemble and hybrid models tended to perform better. The authors noted however that no method found in the literature provides the best results for all evaluation parameters [3].

More recently, Maji et al. proposed a new deep learning architecture that introduces guided decoder and inculcates the properties of Res-UNet and attention gates. Their proposed model was able to outperform the results of the leaderboard models of the BRATS 2019 dataset. It also outperformed its baseline models as well as other current state of the art models. They did, however run into challenges with their proposed model as it was computationally heavy [4].

III. Methodology

There are a few different methods that we plan to use to create an effective model. We plan to add synthetic data, use a multi-model, and use multiple loss functions. The data that we will be adding to our dataset will be created using a diffusion model, likely just stable diffusion. As for incorporating multiple loss functions, we will be combining Dice loss and cross entropy loss, and we will be doing this by calculating both losses and then using some aggregating function (averaging, summation, multiplying, minimum value) to try to optimize.

For this problem we have decided to use the U-Net architecture, a CNN which has already seen success high accuracy in similar problems. This choice of architecture was made for three main advantages which U-Net possesses:

- Fully convolutional (no fully connected layers) making it efficient for varying input sizes which are common to medical imaging.

- Performs well even with limited training data, a limiting feature of our dataset which contains less than 400 examples.
- Fast inference speed, making it suitable for real-time segmentation tasks.

We can modify U-Net to accept multiple channels, and therefore allow us to create our multi-model. The channels that we are interested in using are outlined in the data section. With this in mind, all we should have to do would be to modify the input of U-Net to accept all the inputs that we have.

On top of using U-Net, we will be using other statistical machine learning models to explore any relationships between the a segmentation and the age of the patient, or how long a patient might live. Some features that we might use for these relationships include tumor size, shape, or circularity.

IV. Expected Results

The results that we achieve should be similar to other studies that perform brain tumor segmentation. The metric to use when considering how accurate each segmentation is would be dice score, as it is the standard for segmentation based machine learning methods. From the information that we gathered, we can expect our model training to take a long amount of time, which means that we should start model training as soon as possible. After training is complete, we should be looking dice scores of around 0.7-0.86 [4]. Something to note is that the models were trained on other datasets so there may be some variability with what we can expect, and since we have a different one we could have any dice score, including scores as low as 0.

As for using a segmentation and patient age or other factors, there is not enough literature to conclusively say what accuracy we should be aiming for. We will be removing highly correlated variables beforehand, and performing feature selection to determine what the important factors could be. Accuracy metrics that we should be measuring should be R^2, MAE, MSE, and RMSE.

V. Database

https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation

We will source data from the MICIIA Multimodal Brain Tumor Segmentation Challenge (BraTS2020). This data comes from an international competition focused on the automated segmentation of brain tumors from multimodal MRI scans, and contains nearly 400 training examples of brains all stored in the .nii file format (NifTi). Each example brain is composed of 4 channels of information and a manual segmentation created by Spyridon Bakas Ph.D. and his team.:

- T1 (Native)
- Contrast enhanced T1 (T1CE)
- T2 (Weighted)
- T2 Fluid Attenuated Inversion Recovery volumes (FLAIR)
- Segmentation labels

On top of the image data, we are given a CSV file that contains patient age, survival days, and the extent of resection.

VI. Discussion/Conclusion

With our proposed model, we will be aiming to contribute to the study of identifying and slowing the spread of glioblastoma multiforme tumors. Machine learning is a powerful tool that has the ability to help scientists and doctors around the world prevent the loss of life from cancer. We aim to build a model that is efficient and accurate enough to assist doctors in detecting and monitoring glioblastoma multiforme tumors. By leveraging the potential that CNNs have, our proposed model will seek to push the bounds of current segmentation accuracy and efficiency.

With this in mind, it is important to also consider the challenges that this project may encounter. One such challenge may result from the small sizing of the glioblastoma multiforme tumors. Because the tumors take up such a small area of the brain tissue, there is the potential problem of class imbalance. This could potentially result in a missed or inaccurately segmented tumor region. Another challenge our team may encounter is the overfitting the machine. Because of how much data we are training the machine on, there is a potential chance that the model may fail to accurately identify the tumors in the testing data.

We believe that our proposed model will be highly accurate in tumor segmentation. The CNN U-Net has been a proven aid in identifying this specific type of tumor and we are hopeful that our model can provide a better way of accomplishing the task at hand.

References

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[2] Nelly Gordillo, Eduard Montseny, and Pilar Sobrevilla. State of the art survey on mri brain tumor segmentation. *Magnetic Resonance Imaging*, 31(8):1426–1438, Oct 2013.

[3] Anjali Wadhwa, Anuj Bhardwaj, and Vivek Singh Verma. A review on brain tumor segmentation of mri images. *Magnetic Resonance Imaging*, 61:247–259, Sep 2019.

[4] Dhiraj Maji, Prarthana Sigedar, and Munendra Singh. Attention res-unet with guided decoder for semantic segmentation of brain tumors. *Biomedical Signal Processing and Control*, 71:103077, Jan 2022.

Group Project Confirmation

We, the undersigned, confirm that this assignment is a group project. Each member contributed to the tasks and deliverables as outlined below:

Contributions:

- **Tim Perr:** Methodology and expected results
- **Sam Weiss:** Introduction, database
- **Alex Gore:** Literature Review
- **Tony Garnett:** Literature review and discussion/conclusion

Digital Signatures:

- Tim Perr
- Sam Weiss
- Alex Gore
- Tony Garnett